

HDOC Day-37

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

Algorithm

1. Start

2. Display menu:

- 1 → Convert lowercase to uppercase
- 2 → Reverse string
- 3 → Check palindrome
- 4 → Exit

3. Read the user's choice.

4. If choice = 1:

- Read a string.
- Traverse each character.
- If character lies between 'a' and 'z', subtract 32.
- Display uppercase string.

5. If choice = 2:

- Read the string.
- Find its length.
- Swap first character with last, second with second-last, etc.
- Display reversed string.

6. If choice = 3:

- Read the string.
- Find its length.

- Compare characters from both ends.
- If all corresponding characters are equal → palindrome.
- Otherwise → not a palindrome.

7. If choice = 4:

- Exit the program.

8. Otherwise: Display "Invalid choice".

9. Repeat until choice = 4.

C Program:

```
#include <stdio.h>
```

```
// Function to convert lowercase string to uppercase
```

```
void toUpperCase(char str[])
```

```
{
```

```
    int i;
```

```
    for (i = 0; str[i] != '\0'; i++)
```

```
    {
```

```
        if (str[i] >= 'a' && str[i] <= 'z')
```

```
        {
```

```
            str[i] = str[i] - 32;
```

```
        }
```

```
    }
```

```
    printf("Uppercase string: %s", str);
```

```
}
```

```
// Function to reverse a string
```

```
void reverseString(char str[])
{
    int i, length = 0;
    char temp;

    // Find length of string
    while (str[length] != '\0')
    {
        length++;
    }

    // Reverse the string
    for (i = 0; i < length / 2; i++)
    {
        temp = str[i];
        str[i] = str[length - i - 1];
        str[length - i - 1] = temp;
    }

    printf("Reversed string: %s", str);
}

// Function to check palindrome
void checkPalindrome(char str[])
{
    int i, length = 0, flag = 1;

    // Find length
```

```
while (str[length] != '\0')
{
    length++;
}

// Compare first and last characters
for (i = 0; i < length / 2; i++)
{
    if (str[i] != str[length - i - 1])
    {
        flag = 0;
        break;
    }
}

if (flag == 1)
    printf("The string is a palindrome.\n");
else
    printf("The string is not a palindrome.\n");
}

int main()
{
    char str[100];
    int choice;

    do
    {
```

```
printf("\n----- MENU ----- \n");  
printf("1. Convert lowercase string to uppercase\n");  
printf("2. Reverse a string\n");  
printf("3. Check whether a string is a palindrome\n");  
printf("4. Exit\n");  
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
getchar(); // Removes newline left by scanf
```

```
switch (choice)  
{  
    case 1:  
        printf("Enter a lowercase string: ");  
        fgets(str, sizeof(str), stdin);
```

```
        // Remove newline manually
```

```
        int i = 0;
```

```
        while (str[i] != '\0')
```

```
        {
```

```
            if (str[i] == '\n')
```

```
            {
```

```
                str[i] = '\0';
```

```
                break;
```

```
            }
```

```
            i++;
```

```
        }
```

```
toUpperCase(str);
```

```
break;
```

case 2:

```
printf("Enter a string: ");
```

```
fgets(str, sizeof(str), stdin);
```

```
i = 0;
```

```
while (str[i] != '\0')
```

```
{
```

```
    if (str[i] == '\n')
```

```
    {
```

```
        str[i] = '\0';
```

```
        break;
```

```
    }
```

```
    i++;
```

```
}
```

```
reverseString(str);
```

```
break;
```

case 3:

```
printf("Enter a string: ");
```

```
fgets(str, sizeof(str), stdin);
```

```
i = 0;
```

```
while (str[i] != '\0')
```

```
{
```

```

        if (str[i] == '\n')
        {
            str[i] = '\0';

            break;
        }

        i++;
    }

    checkPalindrome(str);

    break;

case 4:

    printf("Exiting the program...\n");

    break;

default:

    printf("Invalid choice! Please enter 1-4.\n");

}

} while (choice != 4);

return 0;
}

```

Test Case Table:

Test Case	Choice	Input string	Expected Output
1	1	Hello	HELLO
2	1	Computer	COMPUTER
3	2	Hello	Olleh
4	2	CSE	ESC

5	3	Madam	String is a palindrome
6	3	Hello	String is not a palindrome
7	3	level	String is a palindrome
8	4	-	Exiting the program
9	5	-	Invalid choice

Compilation and Execution:

The following test cases has been compiled and executed successfully.

Test Case 1:

```
(kali㉿kali)-[~]
└─$ ./a.out

—— MENU ——
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check whether a string is a palindrome
4. Exit
Enter your choice: 1
Enter a lowercase string: hello
Uppercase string: HELLO
```

Test Case 2:

```
—— MENU ——
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check whether a string is a palindrome
4. Exit
Enter your choice: 1
Enter a lowercase string: computer
Uppercase string: COMPUTER
```

Test Case 3:

```
—— MENU ——
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check whether a string is a palindrome
4. Exit
Enter your choice: 2
Enter a string: hello
Reversed string: olleh
```

Test Case 4:

```
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrome  
4. Exit  
Enter your choice: 2  
Enter a string: CSE  
Reversed string: ESC
```

Test Case 5:

```
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrome  
4. Exit  
Enter your choice: 3  
Enter a string: madam  
The string is a palindrome.
```

Test Case 6:

```
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrome  
4. Exit  
Enter your choice: 3  
Enter a string: hello  
The string is not a palindrome.
```

Test Case 7:

```
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrom  
4. Exit  
Enter your choice: 3  
Enter a string: level  
The string is a palindrome.
```

Test Case 8:

```
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrome  
4. Exit  
Enter your choice: 4  
Exiting the program...
```

Test Case 9:

```
(kali㉿kali)-[~]  
$ ./a.out  
  
—— MENU ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check whether a string is a palindrome  
4. Exit  
Enter your choice: 5  
Invalid choice! Please enter 1-4.
```